Code:

Activities.java

```
enum ActivityType {
   ACADEMIC,
   NON_ACADEMIC
}

enum AcademicActivityType {
   ISA,
   MIDPOINT,
   SEA,
   STUDY_TOUR,
   INDUSTRIAL_VISIT,
   INTERNSHIP,
   PLACEMENT,
   WORKSHOP,
   SEMINAR,
   HACKATHON
}

enum NonAcademicActivityType {
   INFOFEST,
   PLATEAUNIA,
   PLAY_ON,
   SPORTS,
   CULTURAL_PROGRAM,
   FRESHERS,
   FAREWELL,
   SOCIAL_OUTREACH
```

```
}

class Activity implements DepartmentActivity {

    private static int activityCount = 0;

    private String activityName;
    private ActivityType type;
    private int year;

    public Activity(String activityName,
                ActivityType type, int year) {

        this.activityName = activityName;
        this.type = type;
        this.year = year;

        activityCount++;
    }

    // Getter for activity type
    public ActivityType getType() {
        return type;
    }

    // Getter for activity name
    public String getActivityName() {
        return activityName;
    }

    // Getter for year
```

```
    public int getYear() {
        return year;
    }

    @Override
    public void printDetails() {
        System.out.println("Activity: " + activityName);
        System.out.println("Type: " + type);
        System.out.println("Year: " + year);
    }

    @Override
    public void generateReport() {
        System.out.println("Activity report: " + activityName);
    }

    @Override
    public void conductActivity() {
        System.out.println(activityName + " conducted by DCST.");
    }

    public static void displayActivityCount() {
        System.out.println("Total activities: " + activityCount);
    }
}
```

CollegeData.java

```
import java.io.*;
```

```
class CollegeData {

    public static void saveData(String data) {

        try (FileWriter writer =
                new FileWriter("DCST_NIRF_Report.txt", true)) {

            writer.write(data + "\n");
            System.out.println("Data saved successfully.");

        } catch (IOException e) {
            System.out.println("Error saving data: " + e.getMessage());
        }
    }

    public static void readData() {

        try (BufferedReader reader =
                new BufferedReader(
                    new FileReader("DCST_NIRF_Report.txt"))) {

            String line;

            System.out.println("\nSaved DCST Information:");

            while ((line = reader.readLine()) != null) {
                System.out.println(line);
            }

        } catch (IOException e) {
            System.out.println("Error reading file: " + e.getMessage());
```

```
        }
    }
}
```

Course.java

```
import java.util.*;

class Subject {

    private String subjectCode;
    private String subjectName;
    private int credits;

    public Subject(String subjectCode,
                String subjectName, int credits) {

        this.subjectCode = subjectCode;
        this.subjectName = subjectName;
        this.credits = credits;
    }

    public void display() {
        System.out.println(subjectCode + " - "
                + subjectName + " (" + credits + " credits)");
    }
}

class Course {

    private String courseName;
```

```
    private int duration;

    private List<Subject> subjects = new ArrayList<>();

    public Course(String courseName, int duration) {
        this.courseName = courseName;
        this.duration = duration;
    }

    public void addSubject(Subject subject) {
        subjects.add(subject);
    }

    public void displayCourse() {
        System.out.println("\nCourse: " + courseName);
        System.out.println("Duration: " + duration + " years");

        System.out.println("Subjects:");

        for (Subject s : subjects) {
            s.display();
        }
    }
}
```

CustomExpectation.java

```
class InvalidMarksException extends Exception {

    public InvalidMarksException(String message) {
```

```
        super(message);
    }
}

class MarksValidator {

    public static void validateMarks(double marks)
            throws InvalidMarksException {

        if (marks < 0 || marks > 100) {
            throw new InvalidMarksException(
                "Marks must be between 0 and 100."
            );
        }

        System.out.println("Valid marks: " + marks);
    }
}
```